

Digital Forensics in Corporate Investigations: Best Practices

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ITMS – 538: Cyber Forensics

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ABSTRACT

The rise in illegal activity in computer and information systems prompted the development of a new field in computer science called "digital forensics." Computer assaults and cybercrimes are on the rise due to the growing reliance of businesses and society on information technology, as well as the fact that online business and e-commerce have become indispensable components of modern global trade. For a variety of reasons, the legal system, law enforcement, computer forensics, and investigations appear to be lagging behind in their attempts to find and apprehend perpetrators. Key aspects of the administrative implications of forensic computing are covered in the article, along with best practices for the digital forensics' investigation process.

3. INTRODUCTION

3.1. BACKGROUND AND SIGNIFICANCE

A recent area of study in computer science called "computer forensics" focusses on the collection, retrieval, and evaluation of electronic data in order to detect and halt computer fraud, collect and save digital evidence for a criminal investigation, or retrieve data that has been inadvertently erased or destroyed. (Casey, 2011)

Investigators employ computer forensics as a technique or strategy to pinpoint the origin of an assault on computers, data-related resources, and systems. (Casey, 2011)

This paper also discusses some of the most crucial aspects of evidence analysis and computer forensics, including matters pertaining to enforcement and investigations. However, the paper mainly highlights best practices in digital forensics and the consequences for management. The following are the main objectives of computer forensics, as inferred from the above

- Unauthorized activity and actual activities are identified.
- Collecting, processing, storing, and maintaining evidence that might be presented in court.
- To better defend computer systems by applying that knowledge and expertise.

3.2 THE IMPORTANCE OF DIGITAL FORENSICS IN CORPORATE INVESTIGATIONS

Digital forensics is a control that financial institutions create and use. It entails recovering and examining data or information pertaining to cybercrime incidents or suspicions that are discovered or stored in the financial institution's main application system or electronic or digital devices (Casey, 2011).

To analyze facts and evidence relating to a specific cybercrime, a team of professionals and specialists who are knowledgeable with the process and digital devices under investigation conduct digital forensics. Experts in conducting examinations of encrypted data utilizing various forensics software, methods, and methodologies are known as cyber forensic specialists. They can retrieve erased information, break passwords, and more to locate proof of the cybercrime (Garfinkel, 2010). Devices

that could hold digital data or information are investigated as part of the digital forensics' procedure (Jones et al., 2006).

Identification, preservation, assessment, and evaluation of the collected digital evidence are all necessary steps in the digital forensics process (Casy,2011). To determine the underlying cause of a certain cybercrime occurrence, subject matter knowledge is needed to uncover and understand electronic data or information. Finding and preserving digital evidence in its purest form is the goal of digital forensics, which enables pertinent investigative processes and conclusions to be drawn.

Digital forensics is a crucial component of the incident response process for corporations and enterprises. A court of law may request the presentation of digital evidence collected from technological devices. As a result, companies or organizations carry out forensics reviews carefully and conscientiously (Garfinkel,2010).

3.3. SCOPE OF RESEARCH

The primary goal of this study is to investigate the most effective methods for carrying out digital forensics in business investigations. Examining the methods, resources, and strategies used by forensic investigators to find, save, and examine digital evidence in business settings is the goal. Businesses must have strong forensic skills to respond to security breaches, fraud, intellectual property theft, and regulatory infractions as cybercrimes and internal wrongdoing get more complicated.

The paper will concentrate on the optimal procedures for every stage of the digital forensic procedure, including:

1. Acquisition of Digital Evidence: Using write-blockers and imaging tools, the study will describe how to properly acquire forensic evidence without corrupting or changing it (Casey, 2011).

2. Evidence Preservation: In digital forensics, maintaining the integrity of the data that has been gathered is essential. Best practices for handling evidence correctly to prevent contamination or destruction will be described in this section, with an emphasis on upholding the chain of custody(Garfinkel,2010).

3. Analysis and Examination: The presentation will explore forensic analysis methods specific to Linux and Windows settings, including looking at log files, memory dumps, metadata, and file systems (NTFS, ext4). We will also look at best practices for examining artefacts pertaining to system operations, user behavior, and possible malware (Casey,2011).

4. Reporting and Documentation: Recording the results of any forensic investigation is an essential part of the process. To guarantee openness and adherence to legal requirements, the study will demonstrate how researchers should draft precise, understandable, and legally admissible reports that include a description of their methods and conclusions.

This paper will conclude by examining new developments in digital forensics, including the use of machine learning tools for automated data analysis, data handling in cloud environments, and the increasing demand for forensic specialists to be flexible in a workforce that is becoming more dispersed and mobile.

4. LITERATURE REVIEW

4.1 DIGITAL FORENSICS: AN OVERVIEW

Understanding computer forensics is essential if you oversee or manage networks and information systems. Using scientific understanding to gather, examine, and present evidence in court is known as

forensics. Forensics literally translates to "to bring to the court." (Carrier, 2005) Finding and analyzing latent evidence is the main focus of forensics.

DNA evidence retrieved from bloodstains, fingerprints left on a window, and hard drive data are just a few examples of latent evidence.

There is a lack of uniformity and standardization in the courts and business due to the novelty of computer forensics. Because of this, it is not yet acknowledged as a legitimate "scientific" field. (Carrier, 2005)

Computer forensics can be defined as the field that integrates aspects of computer science and law to gather and examine data from networks, wireless communications, computer systems, and storage devices in a manner that may be used as evidence in a court of law. (Baggili, & Memon, 2015)

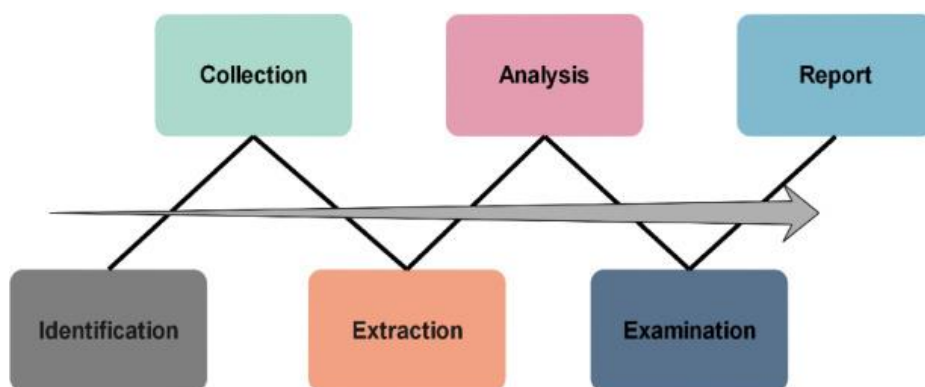


Figure 1 Overview of Digital Forensics

Note. Adapted from Digital Forensics Overview? August 31, 2023

(<https://www.sciencedirect.com/en/resources/what-is-digital-forensics/>). Copyright 2023

by Science Direct.

4.2 TOOLS AND TECHNIQUES USED IN DIGITAL FORENSICS

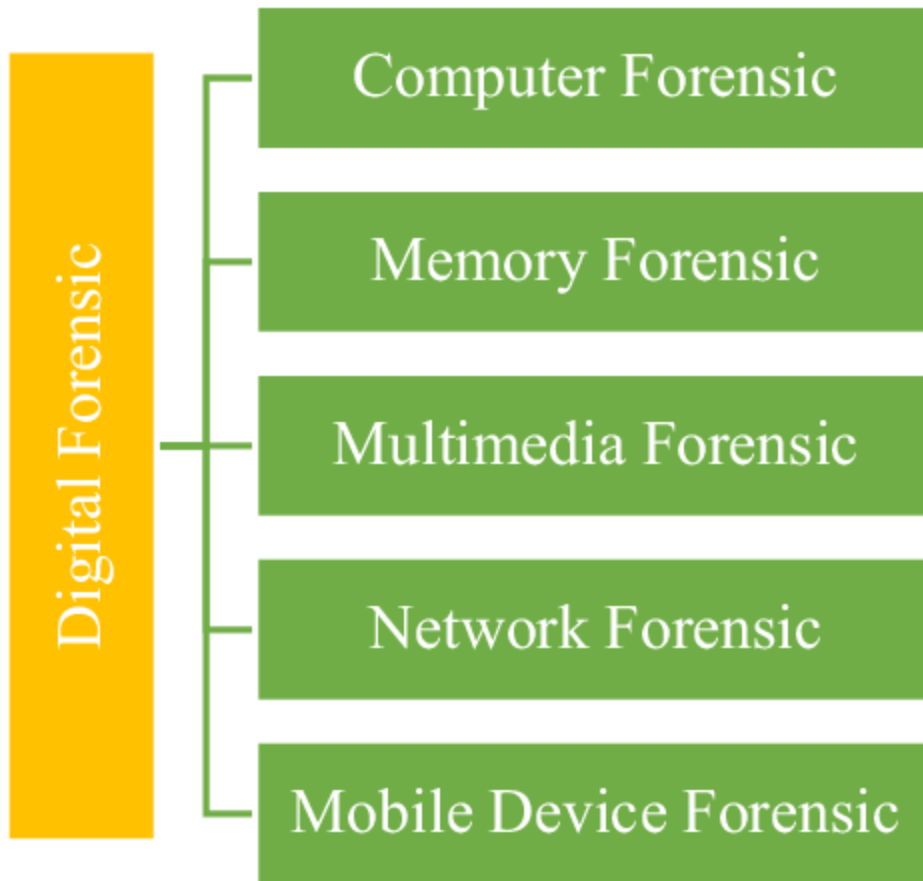


Figure 2 Digital Forensics Tools Classification https://www.researchgate.net/figure/Classification-of-Digital-Forensic_fig1_312559420

COMPUTER FORENSICS

A. Encase

Encase is a tool for gathering data from several devices and using the whole evidence report to find any potential digital evidence. It is a tool that guiding software made and is utilized globally. For the investigative lifecycle, it is used by 80% of US colleges, 75% of power providers, 93% of banks, and 1005 government agencies.

B. The Digital Forensics Framework (DFF)

DFF is an open-source platform with a customizable interface that is utilized by law enforcement agencies, businesses, and educational institutions worldwide. It comes in three versions and options: direct DFF with a supporting license, free DFF, and a paid DFF Pro license. It can read evidence file formats RAW, AFF, and EWF, access distant devices, and retrieve deleted and hidden files. It is used in computer forensics to gather, store, and reveal digital evidence without altering the system or data.

C. Computer Online Forensic Evidence Extractor (COFEE)

The detectives employed this Microsoft-founded forensic program to retrieve digital data from Windows-powered computers. Microsoft works with the National White-Collar Crime Centre (NW3C) and Interpol to develop forensic investigations. It is utilized only by law enforcement and does a live analysis by installing it on a USB stick or hard drive. By inserting the USB pen COFEE into a criminal device, it may start live assessments and gather and analyze evidence in 20 minutes. It features 150 unique tools with a unique graphical user interface (GUI).

D. Forensics Toolkit (FTK)

Access Data Group developed FTK, which can analyze computers, networks, mobile devices, and laptops. Its primary features include quicker search and filtering than competing tools, as well as forensic tools for training and more than 130000 certificates in governance and legal allows to use FTK worldwide. In addition to supporting FAT, NTFS, ext2, and ext3 system formats, it enables responders to get images from criminal systems using specialized equipment (write blockers) and validate obtained photos raw/dd or expert witness known as "Encase".

E. The Sleuth Kit

This is a library that includes a set of command-line tools, supports virtual machines in a new version, and aids in system file analysis, disc image analysis, file recovery associated with an investigation, and SD data recovery.

Using a framework that allows the user to create an automated digital forensic program, the University of Shanghai for Science and Technology Journal (ISSN: 1007-6735) Volume 23, Issue 3, March 2021, Page 372 card is utilized by law enforcement agencies, corporate examiners, and the military.

F. The ProDiscover

This court-backed forensic tool may be used to gather and produce photographs of data, analyze recovered data, and search for recovered data with ease. It can be viewed in two different ways: by group or content. It was created by ARC Group and comes in three different licenses: ProDiscover Basic (\$50), ProDiscover Forensic (\$1679), and ProDiscover Incident Response (\$2,799). It can gather browsing data, device and time zone information, and still preserve original metadata. For example, access time and an incident response version that uses the Connect Collect Protect procedure for real-time analysis during security problems and data breaches.

G. Bulk Extractor

By scanning directories, files, and disc images and storing the extracted contents in another location for automated programs to review and process, this application allows users to extract crucial information without altering system file structures, compatible with Linux, MAC, and Windows. This tool's primary feature is its disregard for file systems.

H. The SIFT (SANS Investigative Forensic Toolkit) SIFT

All digital forensic tools are available on a device running multi-cause forensic; Ubuntu supports a number of digital forensics-related devices and is an open source, free license utility. The Toolkit for

computer forensic investigations utilizing virtual machines handles raw pictures and file systems such as NTFS, HFS, and UFS.

MEMORY FORENSICS

A. Mandiant Redline

This tool is used to analyze memory on a malicious host, gain details about the drivers and processes that are operating in memory, and gather event logs, registry data, network information, system metadata, all services, and active tasks.

B. *Volatility*

This forensic memory framework, which is available for free, is used for incident actions and can extract data from windows crash dumps, including running processes, network ports, registry entries, and DLLs. In Python-based open source software was launched in October 2015 and allows users to construct their own GUI and web interface. They can also add new plugins and update existing ones to create a framework that suits their needs.

C. *Plain Sight*

It collecting digital evidence such as internet histories usage information, analysis ram dumps to extract stored password hashes.

D. Nigilant32

This tool is an incident response tool used for extract (RAM) random access memory content developed by Aigle Risk Management and open source tool to create snapshot reports about live machine working include system, user information system process, network sockets and processes.

E. Memoryze

A free tool developed by Mandiant used to collect, analyze the image of memory was taken, this tool using special batch files for this process using a live system to acquiring image that contains network sockets, system process, and other digital evidence will be founded.

F. Magnet RAM Capture

Specifically made to capture the physical memory of a criminal computer, this forensic tool allows investigators to recover and analyze valuable artefacts found in memory. It can capture data in raw (.DMP/. RAW/.BIN) formats and use Magnet Axion and Magnet IEF to find evidence such as malware intrusion, registry hives, password and username, network connection, decrypted files not found on the hard drive, and support for Windows operating systems up to Windows 10.

G. . HxD (Hex Editor)

Used for Low-level editing capabilities include searching, replacing, exporting checksums, digesting file shredders, concatenating, or dividing files to generate statistics, as well as altering raw discs and random-access memory (RAM).

H. Helix3

Helix3 has a graphical user interface (GUI) for obtaining RAM images and for creating reports that include specific information required for an investigation, such as the operating system version, host information name, IP, and logical drives that are present. The investigator can also use the time and date labelled command to create MD5 hash log files. This tool claims to be based on the Live CD Live Suite, an open source incident response tool that includes a hex editor, password cracking for its commercial version, and the ability to retrieve memory

information like network configuration, services, scheduled jobs, Windows registry chat logs, display snapshots, drivers, and analysis of the data gathered to produce reports.

NETWORK FORENSICS

Digital forensics includes these tools, which sniff live packets. The primary objective is to retrieve digital proof during real-time networks, and these technologies are used to gather evidence from live packet information.

A. The NMAP

This tool is used within the network to map network services, offer services on a computer network such host discovery, and scan ports for network protocols that govern network security. Manually sending certain packets to the target host and examining the source and destination's port and content information are its uses.

B. It is an open source program that supports a wide range of operating systems, including Windows, UNIX, and Linux. It is used to analyze network packets by directly capturing them from the network, filtering them based on user preferences, and displaying information for all the protocols used to transfer the captured packets. It also shows the results in color for each type of packet. Additionally, it may be operated in a mixed mode that is appropriate for collecting multi-protocol packets, including the UDP datagram protocol, the ARP address resolution mechanism, and the TCP transport control protocol. Lastly, it presents the packages together with comprehensive details about them.

C. WinPcap

This tool can be used to capture network packets; it supports IEEE 802.11b/g wireless networks; it can obtain remote packet capture; it can filter packets at the kernel level; it is compatible with Windows OSes; it can capture CTS and RTS control frames; it can capture Probe Response and

request as management frames; and it can capture traffic based on 802.11 data frames that are in charge of authentication.

4.3 CHALLENGES IN CORPORATE INVESTIGATIONS

Five main areas of digital forensics difficulty were identified by Raghavan (2013) based on a review of the field's research:

1. The complexity issue, which necessitates advanced data reduction techniques before analysis, arises from data being obtained at the lowest (i.e., binary) format with growing volume and heterogeneity.
2. The diversity problem arises from both the natural increase in data quantities and the dearth of standardized methods for analyzing and examining the growing quantity and variety of sources, which introduce a multiplicity of file formats, operating systems, and other sorts of data. It is also needlessly complicated for national and international law enforcement organizations to share digital evidence due to the absence of uniformity in digital evidence storage and the structuring of related metadata (Scanlon and Kechadi, 2014).
3. The issue with consistency and correlation brought on by the fact that current technologies are made to identify pieces of evidence, not to support investigations in any other way.
4. The volume issue, which is brought on by growing storage capabilities, the quantity of information-storing devices, and insufficient machinery for analysis.

5. Problems with clock skew/drift, timestamp interpretations, many sources presenting disparate time zone references, and the syntax required to create a single timeline are all part of the unified time-lining dilemma.

Several additional scholars have discovered more specialized difficulties that may be broadly grouped using Raghavan's categorization above. Wazid et al. (2013), Karie and Venter (2015), and Garfinkel (2010) are a few examples. It is generally acknowledged that the amount of data that may be pertinent to investigations is increasing quickly. Between 2003 and 2011, the volume of data per case at the FBI's 15 regional computer forensic laboratories increased 6.65 times, from 84GB to 559GB (Roussev et al., 2013). The increase in storage capabilities over the past few years is one reason for this. Furthermore, the number of devices that need to be examined in a particular inquiry increases due to the growing prevalence of mobile and Internet of Things devices. In addition to the volume of data, using cloud services may make it unclear what data is available and where it is housed. Digital forensic investigations now use wearable and mobile technology more frequently since they have become more commonplace among the general public. The capabilities of these smart devices have advanced over the last ten years to the point that they can now perform on par with the typical home computer, with the main capacity limitations being processing speed and storage. According to Baggili et al. [2015], who have written on forensics on smart watches, this adds to the diversity problem, where a wider range of gadgets are considered candidates for digital forensic analysis. Digital investigations are made more difficult by the many operating systems, file formats, and communication protocols used by mobile and Internet of Things devices. Examining several devices further adds to the consistency and correlation issue, which requires that data obtained from several sources be connected for logical and temporal consistency. The resources of investigators are greatly depleted by the frequent hand execution of this task. The specifications for RAM forensics also

apply in antifoensics scenarios, where a cybercriminal takes steps to prevent evidence from being obtained, for as by developing malware that only lives in RAM. The growing complexity of cybercriminals' operations is another significant obstacle.

Additional problems include the volatility of evidence, the limited lifespan of digital media that could lead to evidence loss, the increasing prevalence of encryption in contemporary communications and data storage, and bandwidth restrictions for transferring data for investigation.

5. ROLE OF DIGITAL FORENSICS IN CORPORATE INVESTIGATIONS

5.1 Preventing Fraud and Misconduct

Financial fraud is a serious danger to businesses since it may result in monetary losses, harm to one's reputation, and legal ramifications. Financial fraud risk may be reduced by utilizing digital forensics, which offers the instruments and methods required to look into, identify, and stop fraudulent activity. Digital forensics techniques and technologies may track and examine network activity, transaction records, and financial data for irregularities and questionable trends that point to dishonest behavior. With the use of data analytics, statistical analysis, and machine learning algorithms, forensic investigators can spot anomalies, disparities, and illicit activities that can indicate possible fraud in financial records.

Using digital forensics, comprehensive investigations may be carried out to find evidence, track transactions, and find the criminals. To find evidence of fraud, forensic specialists search electronic devices, financial documents, and transaction logs using forensic accounting procedures, data recovery tools, and digital evidence analysis methodologies. Digital forensics aids in the preservation and authentication of digital data to back up court cases and law enforcement initiatives against financial fraudsters. To improve staff knowledge of fraud risks, warning signs, and reporting protocols, organizations can use the results of digital forensics to create awareness campaigns and training initiatives.

By helping organizations to efficiently identify, look into, and stop fraudulent activity, digital forensics is crucial for reducing the risk of financial crime. Companies may improve their ability to identify fraud, bolster regulatory compliance, and guard against monetary losses and harm to their brand brought on by fraud incidences. Organizations may protect their resources, uphold stakeholder confidence, and keep a robust and secure financial environment by incorporating digital forensics into their risk management and fraud prevention initiatives. Real-time detection of suspicious activity, illegal access attempts, and insider threats is possible for organizations. By assisting organizations in identifying gaps, vulnerabilities, and weaknesses in their internal controls and security systems, digital forensics aids efforts to manage risk and comply with regulations. Forensic audits, vulnerability assessments, and compliance reviews are ways that organizations might find non-compliance with industry standards, internal rules, and legal obligations. Findings from digital forensics help organizations stop financial fraud and legal infractions by implementing corrective measures, fortifying controls, and improving risk mitigation strategies.

5.2 Data Breaches

Data breaches are a worldwide issue that is growing in frequency. Data breaches were confirmed in 5,258 out of 29,207 reported cyber-attacks, according to the Verizon Data Breach Investigation Report. The first semester of 2022 saw 1,980 instances of data breaches across many industries, a 31.7% rise over the first semester of 2021, according to Flashpoint. According to Accenture, there has been an 11% increase in data breaches since 2018 and a 67% increase since 2014. Furthermore, the enormous impact of data breaches includes both external and internal repercussions for the company. External repercussions impact clients, rivals, suppliers, and other stakeholders. Internal repercussions impact the breached organization's operations, workforce, legal concerns, and finances. For instance, an IBM analysis shows that worldwide damages from data breach events rose from USD 3.86 billion in 2020 to USD 4.24 billion in 2021 and USD 4.35 billion in 2022. Given that Cybersecurity Ventures has estimated that the effect of damages resulting from such catastrophes would exceed USD10.5 trillion by 2025, this figure is expected to continue to rise.

In order to identify data breaches and obtain crucial information about the origin and scope of occurrences, digital forensic investigation is required. Because digital devices might include mobile devices, network devices like routers, and traditional PCs, the investigative method is highly reliant on the kind of device and environment employed. As a result, digital forensics has several branches. Additionally, in order to counteract the ongoing rise of new forms of cybercrime, innovative approaches, new technologies, and flexible investigation process models are required. Collecting evidence to address the 5Ws and How (5WH) questions—what happened, who was involved, and when, where, why, and how an incident occurred—is the most important goal of digital forensics.

5.3 Intellectual Property Protection

In the digital age, theft of intellectual property is a serious issue that impacts governments, corporations, and private citizens. Trade secrets, trademarks, patents, and copyrights are all targets of intellectual property theft since they are products of human ingenuity and creativity. In order to detect, lessen, and stop theft, the specialized discipline of digital forensics looks at and evaluates digital evidence. Modern technologies and methods enable digital forensics specialists to find evidence, recreate events, and offer law enforcement, corporations, and legal organizations insightful information.

Theft of intellectual property refers to the acquisition, use, or exploitation of intangible assets—such as inventions, literary works, creative creations, trademarks, and trade secrets—without the consent of their legitimate owners. Trade secret misappropriation, trademark counterfeiting, copyright infringement, patent infringement, and industrial espionage are some of the ways it might happen. For those who own rights, this problem has serious ramifications, including detrimental impacts on the economy, legal issues, reputation, creativity, and fair competition.

An all-encompassing strategy is required to prevent IP theft, which includes implementing technical protections, enforcing laws, fostering awareness and education, and implementing robust security measures. Cooperation is essential between governmental organizations, law enforcement, business groupings, and foreign partners.

In the coming years, the field of digital forensics in intellectual property (IP) protection is set to undergo major advancements and changes. These developments will be fueled by technological innovations, the ever-evolving cyber threats, and the changing regulatory landscapes. An important prediction revolves around the combination of Artificial Intelligence (AI) and Machine Learning (ML) technologies. These advancements are anticipated to greatly improve the capabilities of digital forensics tools and methodologies.²² This integration will allow for a more streamlined and precise analysis of digital

evidence, advanced prediction of cyber threats, and proactive identification of weaknesses and irregularities in strategies to protect intellectual property.

6. BEST PRACTICES IN DIGITAL FORENSICS INVESTIGATION

6.1 Acquisition of Forensic Evidence

The process of gathering forensically valid data from a digital device, like a computer, smartphone, or other electronic device, is known as acquisition in the field of digital forensics.

In order to get a complete and correct copy of the device data that can be utilized for additional analysis, investigators must make sure that the data is captured in an acceptable manner during this procedure to verify its validity and integrity.

In order to obtain this data, investigators need digital technologies that allow data gathering without contaminating or changing the data's initial condition on the device. Making an identical duplicate of the device data and keeping it in a safe place are usually the steps involved in this process.

Data may be analyzed and used for additional research once it has been obtained. In order to draw conclusions from the data gathered, digital forensic investigators usually use a variety of methods and technologies.

In order to guarantee that the evidence is legitimate and acceptable in court, it is imperative that the data be gathered in accordance with compliance rules.

One of the most important steps in the digital forensic procedure is obtaining digital evidence. It entails compiling information from several sources, including networks, computers, mobile devices, and storage media, all the while maintaining the integrity and preservation of the evidence. These are a few typical techniques for gathering digital evidence.

1. Live acquisition: In this technique, information is gathered from an operational system. Capturing dynamic data, such as active processes, network connections, and system details, is usually part of it. Specialized forensic tools or utilities can be used for live acquisition, which is a non-intrusive process.
2. Creating a bit-for-bit copy, or "image," of a whole storage device, such a hard disc or solid-state drive (SSD), is known as disc imaging. The imaging procedure guarantees the preservation of all data, even hidden or erased files. In order to avoid unintentionally altering the original evidence when imaging, write-blocking devices or tools should be utilized.
3. File-level acquisition: Instead of imaging a complete storage device, it can be essential in some circumstances to collect particular files or directories. The amount of data that has to be processed is decreased while enabling the focused capture of pertinent files. Maintaining the files' integrity and recording their metadata, including timestamps and permissions, are crucial.
4. Acquisition of mobile devices: Portable electronics like tablets and smartphones frequently hold important evidence. Data from these devices is collected using specialized instruments and methods.

Physical acquisition (obtaining a bit-by-bit picture of the device's storage) or logical acquisition (extracting data through the device's operating system) are two possible acquisition techniques, depending on the device and its security measures.

5. Network traffic capture: Gathering and examining network data might yield important evidence in situations requiring investigations using networks. Network monitoring software or tools for packet capture can be used for this. It enables the examiner to pull pertinent information from network packets, spot questionable activity, and analyze communication trends.

6. Cloud-based data acquisition: Acquiring data stored in the cloud for forensic analysis can be required due to the growing usage of cloud services. Tools or APIs that permit legitimate and approved access to customer data are frequently provided by cloud service providers. Requesting data from the cloud provider or utilizing specialized tools to retrieve data from synchronized devices or cloud backups are two possible methods for the acquisition process.

The following best practices must be followed during the purchase process:

- Keep a record of the acquisition procedure, including the date, time, place, and people involved.
- Employ write-blocking hardware or software to stop unintentional alteration of the original evidence.
- Keep the chain of custody intact to guarantee the admissibility and integrity of the collected evidence.
- Verify and validate the obtained data to make sure it is accurate and consistent with the original source.
- To avoid data loss or unintentional changes, make many backups of the collected data.

It is important to remember that obtaining digital evidence necessitates proficiency with appropriate forensic methods. A certified digital forensic specialist should be consulted if you lack the requisite knowledge.

6.2 Preservation of Digital Evidence

There are special issues with digital evidence (DE) preservation that go beyond those with conventional evidence preservation. Digital evidence encompasses any binary data that may be helpful in criminal or other legal investigations and lawsuits. Physical media is where digital evidence is stored by definition, although the content and associated data are frequently more significant than the medium itself.

Frequently, evidence examiners may only access digital material; the physical form of the data may be on cloud storage or law enforcement (LE)-provided media, and it may not always be evident where the data is kept.

Digital storage can fit a remarkably big quantity of data into a tiny physical space. It is becoming more common as a result of social media use, mobile phone use, and the wide range of digital assistants known as the Internet of Things (IoT). Although digital data may be easily altered, there are effective methods for both detecting and preventing alteration.

Here are three ways forensic experts preserve evidence before analyzing it

Drive Imaging

Experts in forensics must first build an image of the evidence before they can analyze it. Experts in forensics use drive imaging to produce an exact, bit-by-bit duplicate copy of the evidence. The following considerations should be kept in mind by forensic specialists while examining this photograph.

Data that can be recovered by forensic specialists may be present on certain evidence-wiped discs. Using forensic procedures, all of the erased files may be retrieved.

Analyzing the original file is a mistake that should never be made. The duplicate image should always be used while examining the evidence.

A write blocker is any piece of hardware or software that makes a legal picture more defensible in court. Forensic specialists can utilize this to make a duplicate image of the evidence for additional examination.

CHAIN OF CUSTODY

Every step taken throughout the transfer procedure on the chain of custody (Coc) must be recorded by forensic specialists when they collect media from the client. These factors make chain of custody documentation crucial.

CoC serves as proof that the picture has been in known custody ever since it was produced.

The legal significance of the photograph and the analysis will be void if there is even the slightest violation of the CoC.

Gaps in procession records, such as any situation in which the evidence was exposed or stored in an unsafe location, might lead to serious issues.

HASHING

SHA1 and MD5 are examples of cryptographic hash values that are produced when forensic specialists make a copy of the evidence for examination. Hash values are significant for the reasons listed below.

- They assist in confirming that the picture is a true copy of the original file and authenticating its integrity.

- The hash values are crucial when that replica is used in court. Hash values will vary if the evidence changes even a little.
- Every time you make any changes to a file on your computer, such as creating or editing a new file, a new hash value will be shown for that file.
- Standard file explorers are unable to read hash values and other metadata. Rather, they require specialized software.

There may be issues in court if the hash values on the picture and the original evidence don't match. It also indicates that the evidence has been altered.

6.3 Analysis and Interpretation of Evidence

The process of evaluating the retrieved data to ascertain its relevance to the case is known as analysis. Analysis may be done on a number of topics, such as ownership and possession, application and file, data concealment, and period. Reviewing the service request, obtaining legal authorization to explore digital evidence, investigation leads, and/or analytical leads may all be necessary for analysis.

Timeline analysis

Timeframe analysis is a valuable tool for figuring out when things happened on a computer system. This information may be used to link computer use to a person or people at the time of the occurrences. To relate files of interest to the periods pertinent to the inquiry, two approaches are available: ■ Examining the time and date stamps included in the file system metadata (such as last updated, last accessed, created, and change of status). To determine when the contents of a file were last altered, one example of this analysis would be to use the latest updated date and time.

Data Hiding Analysis

It is possible to hide data on a computer system. Analyzing data hiding can help find and retrieve such information, which may reveal ownership, knowledge, or purpose.

One way to find such discrepancies is to correlate the file headers with the appropriate file extensions.

The existence of discrepancies might suggest that the user purposefully concealed information.

All password-protected, encrypted, and compressed files are accessible, which might be a sign that an effort was made to hide the information from unauthorized users. The password itself might be just as important as the file's contents.

- Steganography.

- Entering a host-protected area (HPA). When user-generated data appears in an HPA, it might be a sign that data is being hidden.

6.4 Reporting and Documenting of Findings

The cornerstone of every digital forensics' inquiry is documentation and evidence. They offer an unambiguous record of the complete investigation, from the first gathering of digital evidence to the last court presentation. The integrity of the inquiry is guaranteed by well-prepared paperwork and evidence, which also enhances the investigation's overall success. In order to identify cybercrimes, security lapses, and other digital occurrences, digital forensics investigations are essential. The correct documentation and evidence preparation are essential to the precision and legitimacy of these investigations. The next section examines the fundamental best practices for painstakingly recording and gathering evidence in digital forensics investigations.

- Standard Operating Procedures (SOPs): Maintaining Uniformity Throughout Investigations

Create and adhere to precise SOPs that specify how to gather, store, examine, and record digital

evidence. Standard operating procedures (SOPs) for digital forensics are organized rules for managing evidence that guarantee reliable, accurate, ethical, and professional investigations. Investigators preserve their integrity and facilitate the efficient use of evidence in court cases and decision-making by following these rules. They go over important procedures such gathering evidence, adhering to the law, chain of custody, preservation, analysis, documentation, data integrity, tools, quality control, reporting, ethics, and being prepared for cross-examination.

First on the Scene: Maintaining the Evidence's Integrity

In a digital forensics' investigation, the initial responder plays a critical role in preserving the integrity of the evidence and affecting the inquiry's credibility. They guarantee the collecting, documentation, and security of unadulterated evidence. Here, mistakes highlight their importance in a trustworthy investigative basis and endanger accuracy and admissibility in court. Roles and circumstances influence the practices, which might include remote imaging or on-site presence. Different tiers, such as government agencies, business investigators, and law enforcement, have different capacities. Essential actions should be taken by the first person on the scene, such as:

Take Pictures of the Scene: To create a preliminary understanding, take pictures of the computer and the scene. Avoid starting the computer if it is off to avoid changing files that can remove crucial information. If the computer is on, take pictures of the scene and setup before thinking about shutting it off. To image the computer without contaminating the data, connect via USB with the appropriate tools. In order to maintain the disk's precise condition, this entails recording RAM data and mirroring the disc.

Using hashes is crucial for confirming the accuracy of data. Hash the suspect's hard drive using a hashing method before taking a system picture. This will provide a distinct fingerprint that may be used as proof. After imaging, compare hashes and hash the copy.

Evidence Gathering: A RAM picture serves as the starting point for live data collecting, providing information about recent activity. Next, use the appropriate forensic tools to image the hard drive locally. These technologies preserve data integrity by ensuring precise, hash-verified copies.

Device Details: To prevent potential risks associated with power buttons, remove all batteries and power cables before turning off the computer. Make a schematic showing how the cables and connections are arranged. Additionally, for precise identification and documentation reasons, provide all relevant device information, such as model and serial number. A thorough comprehension of the investigation scene is made possible by this methodical technique, which guarantees that the setting can be precisely recreated.

- Legal Hold: Preserving the Accuracy of the Evidence

The creation of a legal hold marks the start of the process of gathering documents and supporting evidence. This important stage guarantees that all possible evidence that has been determined to be relevant to an inquiry is protected and kept in its original condition, guards against its manipulation or destruction, and guarantees that its relevance and authenticity are maintained.

7. LEGAL AND ETHICAL CONSIDERATIONS IN CORPORATE DIGITAL FORENSICS

7.1 Chain of Custody and Legal Admissibility

The most crucial and significant procedure for recording evidence is the "chain of custody," which in criminal and civil law refers to the sequence in which the evidence was handled during a case's investigation. When a chain of custody is needed to demonstrate the legitimacy of the artefact or its unaltered state, it is vital to ascertain the beginning and end of the chain. According to a study by Houck and associates, eligibility is solely impacted by possession breaks that take place throughout the chain of custody period.

The legal authority must be reassured that the evidence is genuine and identical to that which was taken at the crime scene. A key component of properly maintaining the chain of custody is the ability to view the original exhibit. It follows that everyone responsible for its custody should be fully aware of their duty to preserve the evidence and to trace each transit or conservation through adequate documentation, which guarantees the chain of custody's integrity.

Best Steps in the Chain of Custody Process

Collections

Gathering digital evidence is the initial stage in the chain of custody.

Using the right forensic tools to prevent data alteration, thoroughly documenting the scene, and making sure that every step made throughout the collection is documented are all examples of best practices for gathering digital evidence. Write-blockers must be used by collectors when working with storage devices in order to stop data modifications.

Documentation

The integrity of the chain of custody depends on accurate documentation.

Along with the identities of those who handled the evidence, this also includes a thorough documentation of the date, time, place, and manner of collecting. To have a transparent audit trail, all actions involving the evidence must be documented.

Preservation

Crucially important is making sure the evidence is kept safe and undamaged. In order to avoid unauthorized handling, digital evidence should be kept in a controlled environment with limited access. The integrity of the evidence is preserved with the use of tamper-evident packaging and safe digital storage options.

Transfer

Stringent protocols must be adhered to when evidence has to be shared between parties. This entails keeping records of the transfer procedure, making sure that all parties approve the transfer, and transporting the evidence in a secure manner. It is necessary to update the chain of custody documents to reflect each transfer.

Evaluation

In order to preserve the chain of custody, digital evidence needs to be properly examined. It is recommended that analysts utilize forensic software that records every activity made throughout the examination. Additionally, it is their responsibility to make sure that only copies are utilized for examination and that the original material is not moved.

Presentation

When presenting evidence in court, its legitimacy must be demonstrated via a well-defined and recorded chain of custody. Experts in forensics must be able to testify to the protocols used to gather, preserve,

move, and examine the evidence. Thorough records and documentation bolster the validity of the data and conclusions offered.

7.2 Privacy Concerns and Ethical Issues

According to Jones, Bjelica, and Rose (2006), digital forensics is a quickly developing subject devoted to investigating the data on computers and other digital devices. In addition to examining more volatile data like main memory and network traffic, this field mostly studies secondary storage, such as magnetic discs and flash drives, which store data long when the power is switched off. Gathering evidence for criminal and civil legal actions, obtaining information for governments or enterprises, normal system monitoring, and virus analysis are some of its main uses (Greengard, 2012; Matthews, 2010). Digital forensics has several legal applications, including tracking down suspects, confirming or refuting alibis, recording criminal conspiracies, exposing child pornography, demonstrating financial fraud, and identifying copyright infringement.

Regulations and Laws Affecting Digital Forensics

A complicated network of rules and regulations that differ by country control digital forensics. The California Consumer Privacy Act (CCPA) in the US, the General Data Protection Regulation (GDPR) in the EU, and several other national and international legislation are important legislative frameworks. These rules set forth standards for the gathering, use, and preservation of data in addition to protecting people's right to privacy and data security.

Privacy Laws (such as the CCPA and GDPR) and What They Mean

Digital forensics is significantly impacted by privacy legislation like the CCPA and GDPR. The GDPR gives people broad control over their data and places strict restrictions on data processing. In order to

maintain compliance when conducting investigations, forensic investigators need to be aware of certain requirements.

Keeping Individual Privacy Rights and Investigative Needs in Balance

Finding a balance between the necessity for investigation and the right to privacy is one of the main ethical conundrums in digital forensics. Forensic investigators frequently have to balance competing demands, such as the need to find evidence and the need to safeguard private data. It takes thorough evaluation of moral standards, legal obligations, and the possible effects on people's lives to strike this balance.

Models for Ethical Decision-Making in Forensic Investigations

Forensic investigators might use ethical decision-making models as frameworks to help them deal with difficult moral challenges. These models assist practitioners in assessing the moral ramifications of their choices, considering the opinions of all parties involved, and coming to well-informed conclusions that respect moral principles. Common frameworks that offer various strategies for addressing ethical issues are virtue ethics, utilitarianism, and deontological ethics.

Examining Electronic Documents Without Violating Privacy

One difficult but essential component of digital forensics is analyzing digital data without violating privacy.

When gathering pertinent data for the inquiry, forensic professionals must employ analytical methods that protect privacy. This entails applying techniques for anonymization and pseudonymization in addition to concentrating on case-specific data rather than general data sweeps.

7.3 Compliance with Corporate Policies and Regulations

To guarantee that investigations are carried out lawfully, morally, and in a way that safeguards the company and its stakeholders, digital forensics must adhere to corporate laws and regulations (Kessler, 2014). In addition to external regulatory frameworks that specify how certain types of data should be

handled, forensic investigators are required to follow established procedures that control data management, evidence collecting, and analysis (Beckett & Sossin, 2017).

The internal rules that specify how an organization should handle digital evidence are known as corporate policies. With an emphasis on preserving the availability, confidentiality, and integrity of data, these rules frequently mirror the organization's larger security posture and goals (Shinder, 2012). For instance, businesses usually have Acceptable utilize Policies (AUP) that specify how staff members are allowed to utilize company computer and mobile device resources (Greene & Soloway, 2010). During a forensic investigation, these policies make it simpler for investigators to link any inappropriate activities to a user's activity by ensuring that staff members are aware of acceptable behaviors and the repercussions of breaking them (Kessler, 2014).

The Data Retention Policy is another important policy that specifies how long certain kinds of data are retained by the company and when they are safely erased. The availability of evidence in forensic investigations is directly impacted by data retention regulations as investigators must adhere to the guidelines and deadlines established for the preservation of digital information. To preserve the inquiry's credibility, it is crucial to make sure that system logs and email correspondence were kept in accordance with policy if the investigation includes their analysis.

8. CHALLENGES AND FUTURE DIRECTIONS IN CORPORATE DIGITAL FORENSICS

8.1 Overcoming Anti-Forensic Techniques

We can classify anti-forensics strategies that target the entire process or, in certain situations, simply specific steps since the forensic process depends on the interpretation of the data gathered during the

evidence collecting phase. We can forecast if the assault will taint the intended outcome from this starting point of the inquiry. Even without understanding the precise nature of the evidence, forensic investigators may frequently nevertheless show that they were able to work with it.

Data

Techniques in this area have a direct impact on data stored on volatile media. Metadata, the data's location, the data content utilized in the attack, and/or the stolen data that has to be recovered are all included.

Storage

The goal of all storage-related strategies is to make the acquisition procedure as challenging and time-consuming as feasible. Custom RAID arrays are among the greatest examples. Offline disc capture is essentially useless when bespoke arrays are built because investigators won't be able to reassemble them later to recover pertinent data.

Hiding

Keeping hidden

Storing sensitive information in a way that makes it unlikely to be discovered or decrypted is the main objective of concealing strategies. This section covers techniques including in-band or out-of-band concealing and steganography. The main idea behind steganography is to conceal data inside data.

Depending on how much information we wish to keep hidden, several file formats may be employed.

Alternative file streams, file-system journal logs, or reserved, unallocated sectors are the main targets of in-band concealment. Generally speaking, concealed data shouldn't violate the specification's format.

Conversely, out-band hiding can make advantage of slack space near the end of files, sectors designated as bad, the Host Protected Area, or beyond the end of a partition.

Source Elimination

Sometimes the best treatment is prevention. In the case of anti-forensics, this is especially true. The most practical approach is to never create the evidence in the first place rather than expend a great deal of effort trying to hide a trace once it has been created. The usage of live Linux distributions might serve as an excellent illustration. Every action is carried out in real time while running Linux from an external media, such as an optical disc or USB, which doesn't leave any evidence on the actual hard drive. Naturally, this only functions when it is mounted in read-only mode.

Fabrication

Falsifying data is a very cunning tactic. In order to make a victim spend the majority of his time chasing his tail, it aims to overload the forensic analyst with false positives and fraudulent leads. In essence, you make a big mess, then let the forensic analyst tidy it up. For instance, you just need to change as many files on the drive as you can if a forensic analyst is going to use file checksums to attempt to identify an intruder. Additionally, it uses strategies like string decorating, adding known files, and creating fictitious audit trails.

8.2 Advancements in Forensic Technology

Although early digital data formats originally appeared in the late 1970s, the area of digital forensics did not become popular until the 1980s. Around this time, more individuals started buying personal computers, and crimes involving computers started to happen. Digital forensics techniques were initially employed to examine computer systems and gather data for criminal investigations.

The field developed systematic procedures and core techniques for gathering evidence and conducting criminal investigations by the 1990s. As Internet usage increased later in the decade, stronger digital forensic techniques were required to handle escalating problems like identity theft and hacking.

Recent years have witnessed tremendous advancements in forensic science technology, leading to new techniques for evaluating evidence and new instruments for collecting it. These developments have transformed the field of forensic science and increased the precision and effectiveness of criminal investigations. The hardware end presents another difficulty in the current era of new technologies, since the quick growth of networks and devices in global networking and communication has made it challenging to investigate cyberattacks. Digital forensics and cybercrime investigations are increasingly very significant fields for information security, law enforcement, and national security. The topic of cybersecurity might be drastically altered by quantum computing technology, which presents both enormous benefits and quick-emerging dangers. For instance, our communication channels' secrecy may be in jeopardy if optimized quantum computers were developed that could solve complicated problems commonly utilized by encryption methods. Concurrently, the deployment of urban quantum communication networks may offer a framework for the deployment of possibly encrypted messaging protocols in order to circumvent this danger.

Cloud Computing and Big Data

In addition to revolutionizing organizational data management, the dispersed nature of cloud storage has given thieves new ways to hide their activity. More than 60% of freshly created data will be stored on the cloud by 2025, according to IDC (2023). This change forces investigators to adjust to tracking and analyzing data across platforms and jurisdictions.

Key Challenges:

Data fragmentation: Evidence may be spread over servers that are geographically separated, necessitating cooperation from several service providers. Weeks or even months may pass throughout this procedure, greatly increasing the amount of time needed to gather evidence.

Tool Restrictions: Log streams and time-series information are examples of unstructured cloud data that is too large for traditional forensic techniques, which are made for localized data.

Legal Inconsistencies: The EU GDPR and the U.S. CLOUD Act are two examples of conflicting data sovereignty regulations that need case-by-case talks for cross-border evidence retrieval.

Internet of Things (IOT)

Tens of billions of IoT devices, including smart homes, cars, and industrial systems, are anticipated globally by 2025. Digital forensics has both possibilities and problems as a result of its development. For example, in the 2020 Munich Tesla Autopilot case (Az. 370 O 2327/20), investigators used GPS trajectories and vehicle EDR data, including as steering inputs and brake activation records, to recreate accident occurrences. In the end, the court used this evidence from the Internet of Things to establish culpability (German Federal Motor Transport Authority/KBA, 2021). The increasing significance of IoT forensics in court cases is demonstrated by this example.

Tools for e-discovery should advance along with smart device technology and software. The Tesla probe also showed how contemporary e-discovery has evolved beyond conventional data recovery, necessitating the incorporation of over-the-air (OTA) update logs and unique vehicle security mechanisms. Additionally, e-discovery will broaden from conventional data decryption to vendor security software forensics.

8.3 Future Trends in Digital Forensics for Corporate Investigations

Artificial Intelligence and Machine Learning

By automating difficult processes and improving the precision of investigations, artificial intelligence (AI) and machine learning (ML) are transforming digital forensics. AI-powered solutions can swiftly examine enormous volumes of data, spot trends, and spot irregularities that human analysts would overlook. As machine learning algorithms advance, they become increasingly adept at identifying new threats and changing hackers' strategies.

Cloud Forensics

Cloud forensics has emerged as a crucial topic of attention as more businesses move their data and apps to the cloud. Forensic investigators have particular difficulties in cloud systems, including jurisdictional concerns, multi-tenant designs, and data instability. To overcome these obstacles, sophisticated cloud forensic tools and methods are being created, which will allow investigators to effectively gather and examine evidence from cloud platforms.

Blockchain Forensics

Blockchain technology is becoming more popular in digital forensics because of its safe and unchangeable record. Tracking transactions and recognizing digital assets on blockchain networks are the main goals of this new discipline. It is especially pertinent when looking into crimes using cryptocurrencies, such fraud, money laundering, and ransomware attacks.

Virtual Reality and Augmented Reality

The potential of augmented reality (AR) and virtual reality (VR) in digital forensics training and visualization is being investigated. Forensic analysts can improve their abilities by training in immersive

virtual reality environments that mimic real-world situations. AR can help with evidence gathering and analysis by superimposing digital data onto real-world criminal scenes.

Quantum Computing

By speeding up data processing and resolving intricate cryptographic issues, quantum computing holds the potential to completely transform digital forensics. Despite its infancy, quantum computing may help forensic investigators better comprehend cyber occurrences by facilitating more effective data decryption and sophisticated simulations.

Big Data Analytics

Forensic investigators can swiftly handle and examine large datasets thanks to big data analytics. Investigators can find hidden linkages, patterns, and correlations in complicated data structures by utilizing modern analytics technologies. This technology is very useful for incident response and extensive cyber investigations.

9. CONCLUSION

Digital forensics is essential to contemporary corporate investigations because it gives businesses the instruments and techniques they need to identify, examine, and address internal infractions and cybercrimes. An exponential increase in cybercrime, such as fraud, data breaches, and intellectual property theft, has resulted from businesses' growing reliance on digital technology. The collecting, preservation, and analysis of digital evidence have been the main topics of this paper's exploration of the best practices for doing digital forensics inside corporate investigations. Because even little mistakes in data management might make evidence inadmissible in court, the paper emphasized how crucial it is to preserve the integrity of the evidence throughout the investigation. It also covered the important role that

forensic methods and technologies play in helping investigators find deleted or concealed information that might be essential to solving crimes.

Important issues were also looked at, including the growing amount of data, the complexity of contemporary digital ecosystems, and the changing character of cyberthreats. These difficulties highlight the necessity of ongoing innovation and modification in digital forensic techniques. For example, combining artificial intelligence and machine learning may simplify data processing, increasing the precision and speed of forensic investigations.

In summary, digital forensics plays a crucial role in corporate security by assisting companies in navigating the intricate web of cyberthreats. To be ready to tackle the always shifting terrain of digital crime and misbehavior, organizations need to remain up to date on new tools, strategies, and rules as the sector develops. Future technological developments like blockchain, big data analytics, and quantum computing should improve the capabilities of digital forensics and offer even more reliable business investigative options.

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